LAB 4: DATA FLOW DIAGRAM

Health Insurance Claim Portal

Software Engineering and Design Principles - B

Prepared for Lab Submission

KIET • Software Engineering and Design Principles - B • Lab 4

1. Objective of the DFD

The Data Flow Diagram (DFD) represents how information enters, moves through, and leaves the Health Insurance Claim Portal.

The main objectives are:

- To show how a patient or hospital submits a health insurance claim.
- To identify external entities, processes, data stores, and data flows.
- To represent policy verification, document checking, claim assessment, and decision notification.
- To show the system at different levels of abstraction.
- To provide a clear blueprint for understanding the portal's requirements.

System purpose: The portal helps users submit claims, allows the system to verify policy and supporting documents, and communicates the claim decision.

2. DFD Symbols Used

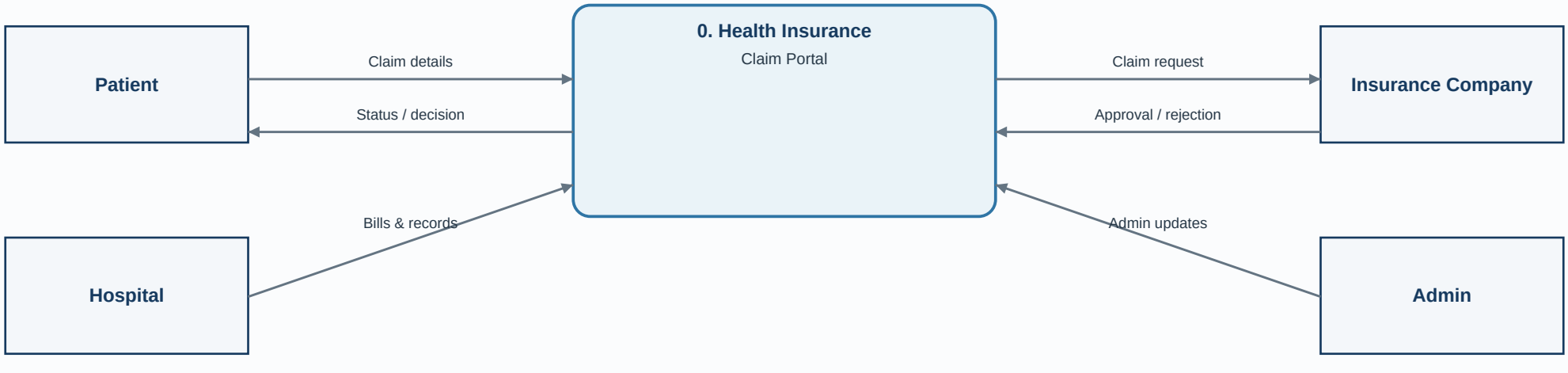
The following standard symbols are used in the diagrams:

Symbol	Name	Meaning
Rectangle	External Entity	A person or organization outside the system that sends or receives data.
Rounded rectangle / circle	Process	Transforms input data into useful output.
Open-ended / parallel-line box	Data Store	Stores data for later retrieval and use.
Arrow →	Data Flow	Shows the direction in which data moves.

Note: Arrows are labelled with the information being transferred, not with actions.

3. Context Diagram (Level 0)

Level 0 shows the complete portal as one process and its interaction with external entities. It does not show the internal steps of claim processing.

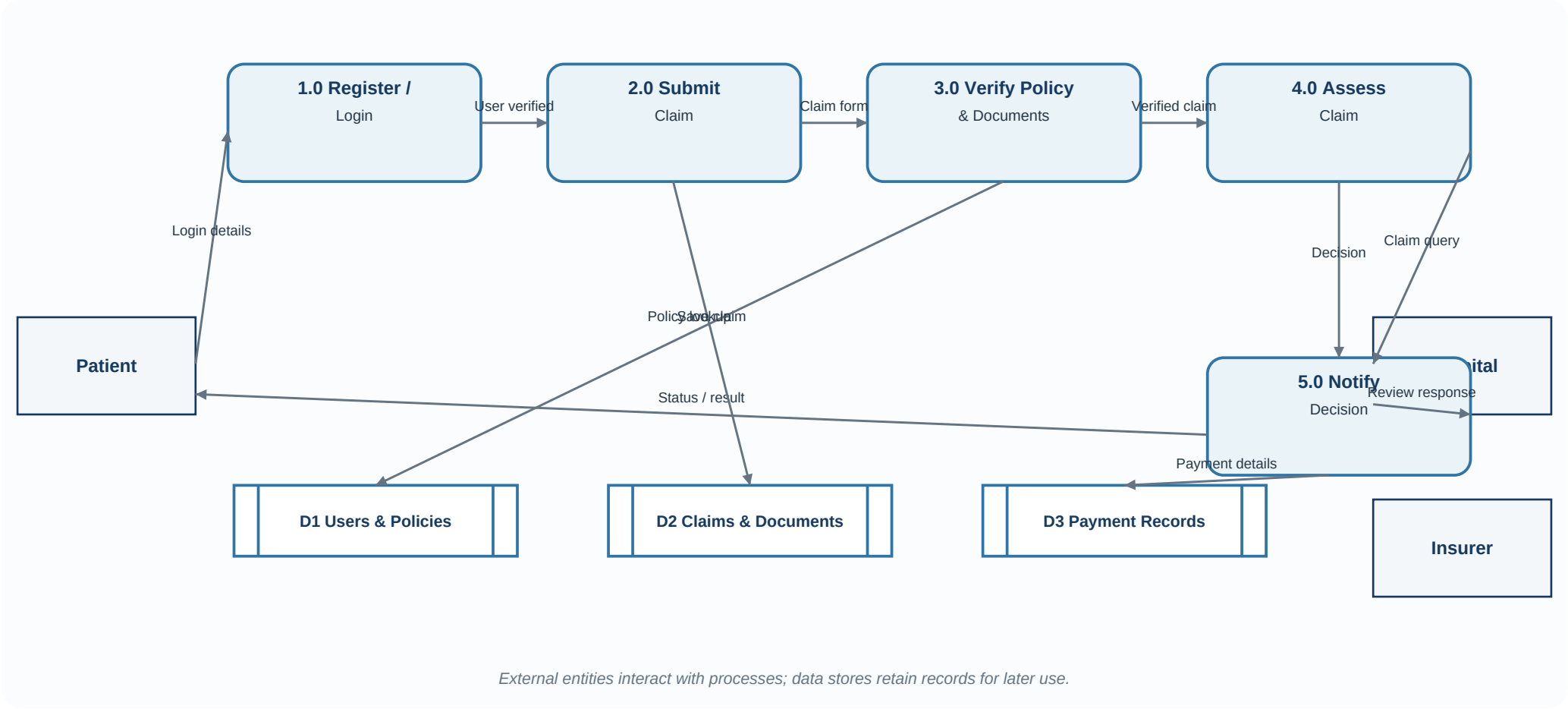


System boundary: all claim information is handled through the portal.

External entities: Patient, Hospital, Insurance Company, and Admin.

4. Level 1 DFD

Level 1 divides the portal into its major processes and shows the main data stores used during claim handling.

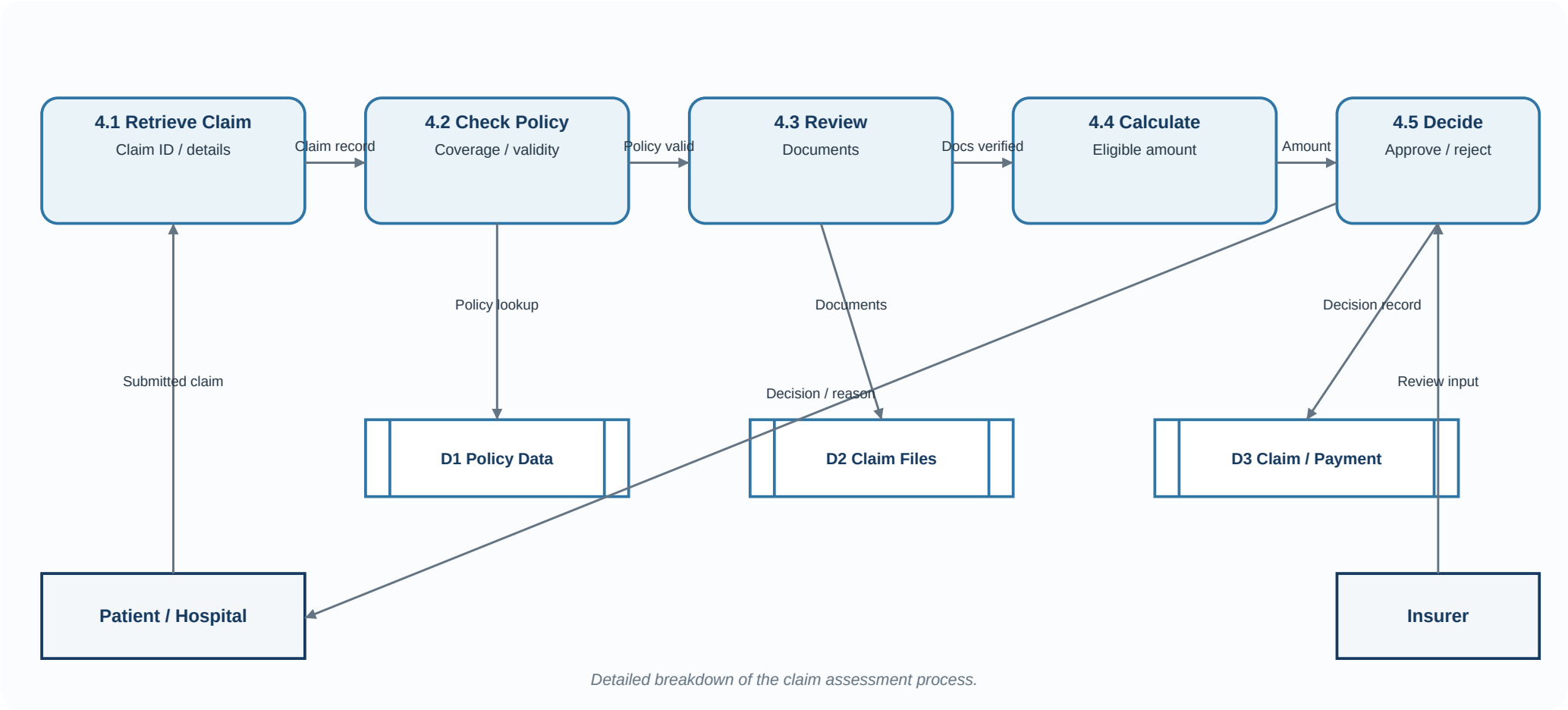


External entities interact with processes; data stores retain records for later use.

Processes: Register/Login, Submit Claim, Verify Policy and Documents, Assess Claim, and Notify Decision.

5. Level 2 DFD – Claim Assessment Process

Level 2 expands Process 4.0 (Assess Claim) into smaller steps to explain how a claim is checked and a decision is prepared.



Detailed breakdown of the claim assessment process.

Output: An approval or rejection decision with the eligible amount or reason for rejection is recorded and communicated.

6. Major Data Flows & Stores

Data / Store	Description
Patient Details	Personal and contact information used for registration and claim identification.
Policy Details (D1)	Policy number, coverage, validity, and insured member information.
Claim Details	Claim form containing treatment, admission, and expense information.
Documents (D2)	Bills, prescriptions, discharge summary, and other supporting records.
Claim / Payment Records (D3)	Claim status, decision, approved amount, and payment or settlement details.
Decision Notification	Approval or rejection status and the reason or next required action.

Important: The data stores are logical representations. The actual portal may use one or more database tables or document storage services.

7. Assumptions

- The patient or hospital provides valid claim information and supporting documents.
- The patient has an insurance policy registered with the insurance company.
- Policy details and coverage information are available for verification.
- The portal checks the submitted documents before claim assessment.
- The insurance company or its authorized workflow makes the final approval or rejection decision.
- The portal stores claim status and decision details and makes them available to the user.
- Payment or settlement is processed according to the insurer's rules and is recorded in the portal.
- Data stores are represented logically; the physical database design may vary.

Scope note: This DFD presents a general health insurance claim workflow. Exact verification rules, approval authority, and payment integration may differ between insurance providers.

8. Conclusion

The DFD clearly represents the flow of information through the Health Insurance Claim Portal.

- Level 0 shows the overall system boundary and its interaction with the patient, hospital, insurance company, and admin.
- Level 1 explains the major activities, including login, claim submission, verification, assessment, and notification.
- Level 2 gives a detailed view of the claim assessment process, from retrieving the claim to recording the decision.
- The data stores show how policy, claim, document, and payment information is maintained for future use.

This DFD helps in requirement analysis, system design, communication, and documentation. It can be refined further when the final portal workflow and database structure are decided.

KIET • Software Engineering and Design Principles - B • Lab 4